RISHOBAN KANDEEPAN

Embedded Firmware Development | Embedded C/C++ | ARM Cortex-M/ESP32 | RTOS & Secure Systems

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Summary

Embedded firmware developer with hands-on experience in Embedded C/C++, microcontroller-based control systems, sensor integration, and hardware-software debugging. Experienced in STM32/ARM Cortex-M bootloader development, firmware validation, flash memory handling, CRC32 integrity checking, and AGV firmware development with PID-based line following, EEPROM storage, and real-time sensor handling. Skilled in ESP32, STM32, ATmega2560, UART, I2C, SPI, GPIO, PWM, ADC, Git, STM32CubeIDE, PlatformIO, and embedded debugging tools. Currently strengthening knowledge in FreeRTOS, Embedded Linux, secure firmware updates, firmware signing, and authenticated embedded communication.

Technical Skills

Languages: C, C++, Python, Assembly; basic Java

Firmware & Low-Level: Bare-metal firmware, bootloader design, flash memory programming, linker scripts, vector table relocation, interrupt-driven logic, timers/PWM, ADC acquisition, EEPROM/NVM storage, sensor calibration, state-machine design, serial debugging

Platforms & Interfaces: ESP32, STM32/ARM Cortex-M, ATmega2560, PIC16F877A, Raspberry Pi; UART, I2C, SPI, GPIO, PWM, ADC, TCP/IP

RTOS, Linux & Security Focus: FreeRTOS task scheduling concepts, Embedded Linux GPIO/I2C/UART interaction, firmware validation, CRC32 integrity checking, secure boot concepts, firmware signing, TLS/HTTPS, certificate validation

Tools & PCB Design: Git, GitHub, VS Code, PlatformIO, STM32CubeIDE, STM32CubeMX, ST-Link, OpenOCD/GDB, KiCad, EasyEDA, Proteus, MATLAB

Experience

Electronic Engineering Trainee, Variosystems – Sri Lanka — June 2025 – Nov 2025

- Developed embedded firmware improvements for an industrial AGV, integrating load-cell calibration, EEPROM-based data persistence, PID line-following, white-zone detection, adaptive turning, and non-blocking timer logic.
- Performed firmware testing, sensor calibration, route validation, and hardware-software debugging to improve AGV reliability under industrial operating conditions.
- Developed a Raspberry Pi-based PCB alignment verification system using Python, Tkinter, and OpenCV, with conveyor/buzzer outputs triggered by real-time orientation validation.

Engineering Trainee, Colombo Dockyard PLC – Sri Lanka — Jan 2023 – April 2023

- Assisted in maintenance and troubleshooting of shipboard electrical and communication systems, supporting operational reliability and safety.
- Contributed to energy-efficiency improvements for welding machines by identifying idle power usage issues and supporting practical optimization work.

Education

Sri Lanka Institute of Information Technology (SLIIT), BSc (Hons) in Electrical & Electronic Engineering — June 2021 – Present

- **Relevant Coursework:** Embedded Systems Programming, Digital Design, Electronic Design, Robotics and Control, Internet of Things, Computer Vision and Image Processing

University of Moratuwa, Bachelor of Information Technology (BIT) – First semester — Feb 2026 – Present

Royal College, Colombo, G.C.E. Advanced Level and Ordinary Level completed — Jan 2012 – Oct 2020

Projects

Secure STM32 Bootloader & Firmware Update System – ARM Cortex-M Firmware Security GitHub

- Developed a custom STM32 Blue Pill bootloader with relocated application execution, flash erase/write handling, update flag mechanism, and firmware update workflow.
- Verified bootloader behavior using LED-coded status indicators, ST-Link breakpoints, watch variables, STM32CubeProgrammer flash memory inspection, and ESP32-C3 UART bridge testing.
- **Tools:** STM32 Blue Pill, Embedded C, STM32CubeIDE, ST-Link, Flash Memory, Linker Script, CRC32, Python

Autonomous Guided Vehicle (AGV) – Embedded Firmware & Sensor Integration GitHub

- Developed embedded firmware for an industrial AGV with load-cell calibration, EEPROM persistence, PID-based line following, white-zone detection, adaptive turning, pushbutton override, OLED feedback, and non-blocking timer logic.
- Performed sensor calibration, route validation, and hardware-software debugging to improve navigation reliability under industrial operating conditions.
- **Tools:** Embedded C/C++, ATmega2560, CZL601 Load Cell, IR Sensor Array, EEPROM, OLED, Motor Control

ESP32 Custom Development Board – PCB Design & Embedded Hardware GitHub

- Designed a custom ESP32-based development board for embedded prototyping, including power regulation, boot/reset circuitry, GPIO breakout, programming interface, and sensor/IO expansion.
- Created schematic, PCB layout, and fabrication files while considering component placement, signal routing, power distribution, and debugging accessibility.
- **Tools:** ESP32, KiCad, PCB Design, Schematic Design, Power Regulation, GPIO, UART/I2C/SPI

Wearable Kyphosis Monitoring System – ESP32 Sensor Acquisition & Feedback GitHub

- Developed an ESP32-based wearable monitoring system using EMG and gyroscope sensors for real-time posture-related data acquisition and feedback control.
- Implemented sensor data reading, noise filtering, serial data output, and vibration feedback using micro DC vibration motors.
- **Tools:** ESP32, Raspberry Pi 5, MyoWare Sensor, MPU6050, Embedded C/C++, Python

Vision-Based PCB Alignment Verification – Raspberry Pi Automation GitHub

- Developed a Raspberry Pi-based PCB alignment verification system using Python, OpenCV, Tkinter, ORB feature matching, and real-time ROI selection.
- Integrated orientation validation with conveyor/buzzer outputs to support automated inspection under rotation, scale variation, and partial occlusion.
- **Tools:** Raspberry Pi 5, Python, OpenCV, Tkinter, GPIO

Certifications & Professional Development

Selected Certifications: Advanced BSP Development with Embedded C; Advanced Architecture in Embedded Software Design; Foundations of Embedded Software Design; Embedded Systems using C

Current Development Focus: FreeRTOS, Embedded Linux, authenticated firmware update, firmware signing, TLS/HTTPS communication, certificate validation, and secure embedded communication.

Volunteering

Robofest 2023 & 2024 – Supported outreach activities for a nationwide robotics competition, including student guidance on line-following robot concepts.

Referees

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